

Settley: Conditional Liquidity Infrastructure for Illiquid Assets

Temisan Gerrard
Independent Researcher & Founder, Settley
temisan@settley.co

Draft Version 3.0 — June 2026

Abstract

SETTLEY is proposed conditional liquidity infrastructure for tokenized ownership interests in inherently illiquid assets. The protocol starts with real estate, but the underlying problem is broader: tokenization can represent ownership, yet it does not by itself create credible exit, collateralization, settlement, or recovery. Across real estate, infrastructure, private credit, natural resources, and alternative assets, investors face the same structural constraint: assets may be valuable and cash-flowing, but secondary liquidity is thin, episodic, gated, or unavailable.

SETTLEY addresses this gap by separating issuance from liquidity and credit. The proposed architecture combines tokenized asset ownership, productive asset-specific vaults, stale-but-verifiable NAV oracles, dynamic discounting, compliance-gated transfer, ownership-backed borrowing, and legally defined recovery or wind-down mechanisms. Each vault functions as a programmable credit facility attached to a specific illiquid asset or asset pool. Liquidity is explicitly conditional rather than guaranteed: vaults may charge availability fees, lend against ownership interests with collateral haircuts, buy interests at a discount to NAV, or combine these approaches, subject to committed capital, utilization limits, compliance checks, valuation staleness, and circuit breakers. The protocol therefore prices illiquidity instead of pretending to eliminate it.

Real estate is presented as the Phase 1 implementation because it is large, familiar, collateralized, and visibly illiquid. The longer-term thesis is asset-agnostic: SETTLEY is infrastructure for converting illiquid ownership into programmable liquidity and ownership-backed credit markets, while preserving legal enforceability and explicit failure states.

Disclaimer. This paper is a protocol and legal-architecture proposal. It is not an offer to sell securities, investment advice, a legal opinion, evidence of regulatory approval, or evidence of realized investor demand. All ownership, transfer, custody, redemption, lending, collateral, forced-transfer, and wind-down mechanics described here are subject to governing documents, regulatory clearance, and applicable law.

AI Disclosure. The author used AI-assisted drafting and editing tools to revise structure, wording, and formatting. The author reviewed and accepts responsibility for the paper’s claims, citations, analysis, and final text.

1 Introduction: Issuance Is Solved; Liquidity Is Not

Tokenization has made it easier to represent ownership claims on-chain. It has not solved liquidity.

Most real-world asset (RWA) tokenization projects focus on issuance: how to place shares, fund interests, property interests, receivables, royalty rights, or other economic claims into a programmable wrapper. That work matters, but it addresses only the first layer of the problem. A tokenized claim over an illiquid asset remains illiquid if there is no credible balance sheet willing to buy it, lend against it, price its risk, or recover value when the asset underperforms.

This distinction is central. Issuance creates representation. Liquidity requires market structure.

Public markets solve liquidity through continuous trading, market makers, clearing systems, margin rules, disclosure regimes, and institutional depth. Illiquid asset markets do not have those conditions. Real estate, infrastructure assets, private credit, natural resources, royalty streams, equipment leases, and other alternative assets are often valuable and cash-flowing, but they trade through slow legal processes, private

negotiations, fund redemptions, securitization windows, or manager-led exits. Their valuations are episodic. Their cash flows are asset-specific. Their transferability is legally constrained. Their recovery depends on enforceable documents, not only software.

SETTLEY begins from the premise that illiquidity is not a defect to be wished away. It is a risk to be priced, allocated, and governed.

The protocol proposes conditional liquidity infrastructure for illiquid ownership interests. Each asset or asset pool can be paired with a dedicated vault that operates as a productive asset-specific balance sheet. Liquidity providers (LPs) commit capital to that specific exposure and may earn availability fees for undrawn commitments, loan interest from secured credit, discounted ownership upside from acquired positions, and asset-level yield where applicable. Investors seeking liquidity may borrow against their ownership position, sell tokens to the vault at a disclosed discount to the last verified NAV, combine borrowing and partial sale, or use the facility for bridge financing, asset improvement finance, construction support, or working-capital needs.

This architecture reframes tokenization. The goal is not to make a building, bridge, loan pool, royalty stream, or energy project trade like a liquid public equity. The goal is to make ownership more financeable by defining the conditions under which liquidity exists, the price at which it is available, the collateral rules governing credit, the compensation owed to committed liquidity providers, and the recovery process when liquidity fails.

Real estate is SETTLEY's first implementation because it is the clearest instance of the problem. It is large, income-producing, legally familiar, and structurally illiquid. Savills estimates the total value of global real estate at roughly USD 300–380 trillion [1]. A Kenya diaspora real-estate proof of concept can test the protocol in a constrained, observable environment. But real estate is not the thesis. The thesis is that illiquid ownership can become programmable financial collateral when liquidity is conditional, priced, compliant, and recoverable.

2 The Illiquidity Problem Across Global Asset Markets

Illiquidity is not unique to real estate. It is a defining feature of many of the world's largest and most economically important asset classes. These markets are not illiquid because they lack value. They are illiquid because ownership claims are legally complex, valuations are episodic, transaction costs are high, buyer pools are specialized, and recovery requires asset-specific enforcement.

Tokenization addresses the representation of ownership, but it does not by itself solve these constraints. A token can make a claim easier to record and transfer. It cannot guarantee that a buyer exists, that a lender will accept the token as collateral, that NAV is current, or that liquidation proceeds will be available on demand. This is the gap SETTLEY is designed to address.

2.1 Real Estate

Real estate is the most familiar form of illiquid ownership. Direct property sales can take months. Fractional property interests are harder still: there may be no natural buyer, no standardized secondary venue, and no mechanism to exit without selling the whole asset. Public REITs solve liquidity by pooling assets into listed securities, but investors lose direct asset-level exposure. Non-traded REITs and private real-estate funds often provide only limited repurchase programs or manager-controlled redemption windows [2, 3]. Tokenized real-estate platforms improve access and transfer mechanics, but secondary liquidity remains dependent on market depth and buyer demand [4, 5].

SETTLEY's contribution is to make asset-level liquidity and credit explicit. A property token should not be called liquid merely because it can transfer. Liquidity should be measured by committed facility capacity, availability terms, borrowing haircuts, acquisition discounts, facility utilization, freshness of NAV, and the legal recovery path if facility capacity fails.

2.2 Infrastructure Assets

Infrastructure assets such as toll roads, bridges, ports, airports, energy grids, solar farms, wind farms, and data centers are long-duration, high-value, locally regulated, and often cash-flowing. Ownership inter-

ests usually trade through private transactions, fund interests, project-company sales, refinancing, or listed infrastructure vehicles. Secondary liquidity exists, but it is largely institutional, negotiated, and diligence-heavy [6, 7].

Conditional liquidity infrastructure could allow partial interests in infrastructure assets or project companies to support asset-specific exit and credit mechanisms. A vault could be sized to the asset’s concession terms, revenue volatility, regulatory risk, counterparty quality, and recovery rights. Discounts and lending haircuts would price the fact that a bridge or solar farm cannot be liquidated like a public security.

2.3 Private Credit

Private credit includes direct lending, SME lending, invoice finance, trade finance, equipment finance, and asset-backed lending. It is often cash-flowing and collateralized, but the loans themselves are not broadly liquid. Liquidity is typically handled through scheduled amortization, warehouse facilities, securitization, fund-level subscription and redemption mechanics, loan portfolio sales, or private-credit fund secondaries.

SETTLEY-style infrastructure could create loan-pool-specific liquidity and credit facilities. Draw pricing, borrowing haircuts, and exit pricing could reflect delinquency, weighted average life, collateral coverage, borrower concentration, and facility utilization. Recovery could map to collections, collateral enforcement, invoice assignment, trade receivable payment, or loan portfolio sale.

2.4 Natural Resources

Natural-resource assets include mining concessions, timber rights, carbon projects, water rights, and royalty streams. These assets may have significant economic value, but their liquidity is constrained by permitting, verification, environmental obligations, commodity exposure, political risk, harvest cycles, and registry rules. Ownership interests in a concession, forestry project, or carbon project are far less liquid than the commodity or credit they may ultimately produce.

Conditional liquidity infrastructure would not make these assets instantly liquid. It would make the liquidity and credit terms explicit. A vault might lend against verified royalty streams, provide project bridge finance, buy discounted project interests, or provide limited exits from a diversified natural-resource pool. The discount and haircut engine would price permitting risk, valuation staleness, commodity exposure, verification data, and recovery time.

2.5 Alternative Assets

Alternative assets such as music catalogs, film royalties, litigation finance, aircraft leases, equipment leases, sports rights, intellectual property, and data centers often have identifiable cash flows but limited secondary liquidity. A catalog owner may receive royalty income but cannot easily borrow against a small fractional interest. A litigation finance investor may have a claim on proceeds but no predictable exit date. An aircraft lease produces contractual payments, but liquidity depends on lessee credit, aircraft type, jurisdiction, repossession rights, and residual value.

These markets demonstrate why issuance is insufficient. A tokenized royalty interest or lease participation still requires valuation, credit rules, liquidity funding, and recovery. SETTLEY’s asset-specific vault model offers a framework for turning these ownership claims into financeable instruments without misrepresenting them as continuously liquid assets.

2.6 The Common Structure

Across these markets, the same pattern appears:

- the asset can be valuable without being liquid;
- ownership can be represented without being financeable;
- transferability does not guarantee exit;
- NAV is useful but stale;

- buyers demand a discount for immediacy;
- lenders require collateral haircuts;
- recovery depends on legal rights and time;
- liquidity fails if it is not explicitly funded;
- LP economics are stronger when committed capital can earn availability fees, loan interest, discounted acquisition upside, and asset-level yield rather than waiting passively for exits.

SETTLEY should therefore be understood as market infrastructure for this common structure. It does not eliminate illiquidity. It converts illiquidity into an observable, priced, governed condition.

3 Settley Thesis: Conditional Liquidity Infrastructure

SETTLEY’s thesis is that the binding constraint in tokenized RWAs is no longer the mechanical issuance of tokens. Issuance can be solved through legal wrappers, transfer agents, security-token standards, SPVs, fund interests, and smart-contract registries [8, 9, 10]. The unsolved problem is liquidity: who provides it, at what price, under what compliance constraints, against what collateral, and with what recovery path.

The protocol’s contribution is the proposed integration of six primitives:

1. tokenized asset ownership;
2. asset-specific liquidity and credit vaults;
3. stale-but-verifiable NAV oracles;
4. dynamic discount and haircut engines;
5. compliance-embedded transfer and lending controls;
6. legally defined recovery and wind-down mechanisms.

None of these primitives alone is novel. NAV facilities, repo, margin lending, secured credit, private secondaries, asset-backed loans, and tokenized securities all exist in different forms [11, 12]. The design claim is that illiquid tokenized ownership requires these primitives to be combined into asset-specific programmable facilities, with explicit limits and failure states.

4 Universal Architecture

4.1 NAV Oracle

Every illiquid asset requires a valuation reference. The oracle should not pretend to create real-time pricing where no real-time market exists. It should publish episodic, auditable, stale-but-verifiable NAV based on the asset class:

- appraisals for real estate;
- discounted cash-flow and concession valuation for infrastructure;
- loan marks, delinquency data, and collateral reports for private credit;
- reserve reports, offtake contracts, or verified issuance data for natural resources;
- royalty, lease, or litigation cash-flow models for alternative assets.

The oracle is not a fully decentralized price feed in the early design. It is a governed valuation process with on-chain publication, audit trails, appraiser or valuation-agent accountability, emergency impairment rules, and disclosed staleness penalties.

4.2 Asset-Specific Liquidity and Credit Vault

Liquidity must be scoped to the asset or asset pool. A vault backing one asset should not silently back another unless the documents explicitly create cross-collateralization. This prevents platform-wide liquidity claims from hiding asset-level risk.

The vault should be described as a productive asset-specific balance sheet rather than a dormant redemption reserve. It has three primary functions:

1. **Availability liquidity:** LPs commit capital to the asset vault and earn an availability fee for making liquidity available, whether or not it is immediately drawn.
2. **Ownership-backed lending:** token holders borrow against their ownership positions, and LPs earn interest from secured loans.
3. **Discounted acquisition:** token holders seeking full exit sell to the vault at a dynamic discount to NAV, giving LPs potential discounted ownership upside.

This framing supports sell-to-vault exits, borrow-against-ownership transactions, hybrid liquidity, bridge financing, asset improvement finance, construction support, and working-capital facilities. The vault is therefore not just a secondary-market backstop. It is a programmable credit facility attached to an illiquid asset.

4.3 Dynamic Discount and Haircut Engine

The discount and haircut engine is the core economic primitive. It converts illiquidity into a price. For acquisitions, the vault buys at a discount to NAV. For loans, the vault lends at a loan-to-value ratio that incorporates collateral haircuts. Both parameters should account for:

- base illiquidity risk;
- facility utilization;
- NAV staleness;
- asset quality;
- legal enforceability;
- cash-flow volatility;
- concentration;
- recovery horizon;
- macro stress.

For an exit request for asset i , a simplified acquisition price can be expressed as:

$$\text{Exit Price}_i = \text{NAV}_i \times (1 - d_{\text{base},i} - d_{\text{util},i} - d_{\text{stale},i})$$

For a secured loan against asset i , a simplified maximum advance can be expressed as:

$$\text{Advance}_i = \text{NAV}_i \times \text{LTV}_i$$

where:

$$\text{LTV}_i = \text{LTV}_{\text{base},i} - h_{\text{util},i} - h_{\text{stale},i} - h_{\text{risk},i}$$

The acquisition discount and lending haircut are related but not identical. Lending should generally be more conservative than acquisition pricing because the vault must account for enforcement delay, collateral impairment, and liquidation cost.

4.4 Compliance Layer

Most tokenized ownership interests in cash-flowing real-world assets will sit inside securities, fund, lending, property, commodities, or payments regulation. A universal compliance layer should control issuance, transfer, custody, redemption, lending, collateral pledge, liquidation, and disclosures through credentials, jurisdiction rules, sanctions checks, suitability checks, and transfer restrictions.

The contemplated design uses non-transferable compliance credentials, such as soulbound tokens or verifiable credentials, to represent KYC, AML, accreditation or suitability status, jurisdiction, sanctions status, and credential expiry [13, 14, 15]. The token, vault, and lending contracts verify these credentials before transfer, pledge, borrowing, acquisition, or liquidation. Code-level checks supplement, but do not replace, legal documents and regulated onboarding.

4.5 Recovery and Wind-Down Mechanism

Illiquid assets require a defined recovery path. The protocol cannot stop at transfer restrictions, redemption queues, or lending contracts. It must explain what happens when liquidity is depleted, a borrower defaults, NAV is impaired, or the asset fails.

Recovery may involve:

- sale of property;
- sale of project-company interests;
- collection or sale of loans;
- enforcement of receivables;
- concession transfer;
- royalty assignment;
- collateral liquidation;
- lender step-in rights;
- asset wind-down and final distribution.

Recovery is asset-specific and legal. Smart contracts can automate notices, freezes, collateral transfers, and distribution logic, but enforceability ultimately depends on governing documents, statutory rules, creditor priority, courts, regulators, and local asset law.

4.6 Governance Layer

Each asset requires governance over NAV updates, emergency impairment, facility parameters, lending limits, use-of-proceeds rules, major asset actions, recovery triggers, manager replacement, and wind-down. The governance system is universal; the voting thresholds and legal consequences are asset-specific.

Table 1: Universal and Asset-Specific Components

Component	Universal Function	Asset-Specific Variation
NAV Oracle	Publishes stale-but-verifiable valuation reference	Appraisal, DCF, loan mark, reserve report, royalty model
Vault	Provides asset-specific liquidity and credit capacity	Reserve ratio, facility size, availability fee, loan purpose, eligible draws
Discount / Haircut Engine	Prices immediacy and collateral risk	Asset class, staleness, utilization, recovery horizon
Compliance Layer	Controls issuance, transfer, pledge, lending, liquidation	Securities, lending, commodities, property, fund, or local law
Recovery Mechanism	Defines enforcement and wind-down	Property sale, receivable collection, concession transfer, royalty assignment
Governance Layer	Controls thresholds and emergency actions	Voting standards, lender rights, asset-operator replacement, sale authority

5 Vault as Productive Asset-Specific Balance Sheet

The vault should not be presented as a passive reserve that waits for investors to sell. That model understates both the architecture and the LP proposition. A better description is:

The vault is a programmable credit facility attached to an illiquid asset.

Under this framing, LP capital is committed to an asset-specific balance sheet that can support liquidity, secured lending, and asset-level financing. The vault does not guarantee exits. It defines the conditions under which capital is available, the price of that capital, the collateral rules that protect LPs, and the recovery process if the asset or borrower underperforms.

5.1 Availability Liquidity

LPs commit capital to the asset vault before liquidity is drawn. In exchange, they may earn an availability fee for standing ready to provide liquidity to that asset or asset pool. This solves a weakness in a pure redemption-reserve model, where LP capital may sit idle until an exit request appears.

Availability fees compensate LPs for reserving balance-sheet capacity for a specific illiquid asset, accepting asset-level concentration risk, remaining ready to fund exits or loans, bearing the opportunity cost of committed capital, and supporting the asset’s perceived financeability. The fee can be paid from protocol fees, asset cash flows, borrower commitment fees, issuer fees, or a combination, depending on the asset class and offering documents.

5.2 Ownership-Backed Lending

The vault can lend against tokenized ownership positions. A token holder pledges their ownership interest and receives a secured loan based on NAV, collateral quality, and the asset’s recovery profile. LPs earn interest and fees from the loan rather than waiting for a holder to sell.

This use case is central because many holders do not want to exit ownership permanently. They may need temporary liquidity, bridge capital, or working capital while preserving upside exposure. Lending gives the holder liquidity without forcing a discounted sale.

5.3 Discounted Acquisition

If a holder wants full exit, the vault can buy the ownership position at a dynamic discount to last-verified NAV. In this case, LP capital becomes buyer capital. LPs receive potential upside from acquiring ownership exposure below NAV, plus asset-level yield where applicable.

The acquisition discount should reflect base illiquidity risk, NAV staleness, vault utilization, asset operating performance, legal enforceability, recovery horizon, and LP concentration risk.

5.4 Supported Liquidity Modes

The balance-sheet framing supports a broader set of transaction types:

- **sell-to-vault:** full or partial exit through discounted acquisition;
- **borrow-against-ownership:** secured loan collateralized by tokenized ownership;
- **hybrid liquidity:** partial sale plus secured borrowing;
- **bridge financing:** temporary liquidity before refinancing, sale, rent collection, project milestone, receivable payment, or distribution;
- **asset improvement finance:** funding for repairs, capex, tenant improvements, energy upgrades, or operational improvements;
- **construction or working-capital support:** credit lines tied to project budgets, draw schedules, covenants, and milestone verification.

5.5 LP Return Stack

The LP proposition becomes stronger under this model. LPs are not merely waiting for redemption demand. They can earn:

- availability fees for committed but undrawn capacity;
- loan interest from ownership-backed credit;
- origination, commitment, or draw fees where legally permitted;
- discounted ownership upside when the vault acquires interests below NAV;
- asset-level yield attached to acquired positions;
- potential seniority or collateral protection in lending structures.

The vault’s economic function is not only to absorb exits. It is to make illiquid ownership financeable by attaching a dedicated credit balance sheet to the asset.

6 Ownership-Backed Liquidity

The current tokenized-asset model usually assumes that liquidity means sale. An investor owns a tokenized interest, wants liquidity, and sells that interest to another buyer or to a protocol vault. That pathway is important, but incomplete.

In mature financial markets, owners often obtain liquidity without selling the underlying asset. Home-owners borrow through home-equity loans and home-equity lines of credit. Public-market investors borrow through margin lending. Securities dealers finance inventories through repo. Private funds use NAV facilities secured by portfolio value. Asset managers use subscription lines, borrowing-base facilities, and secured lending structures to create liquidity against ownership claims or expected cash flows [11, 12].

SETTLE incorporates the same principle: tokenized ownership can support both sale liquidity and credit liquidity.

6.1 Pathway 1: Sell-to-Vault

The vault can support outright purchase. A verified investor sells tokens to the asset-specific vault at a discount to last-verified NAV. LPs provide the capital. In return, the vault or LP tranche receives the ownership exposure and any associated cash flows. This is economically similar to a market maker or secondary buyer demanding compensation for immediacy and illiquidity risk.

This pathway is appropriate when the investor wants full exit, the vault has sufficient committed capacity, the asset’s NAV is recent enough, LPs are willing to acquire the exposure, compliance rules permit transfer, and the discount adequately compensates the buyer.

6.2 Pathway 2: Secured Lending Against Ownership Interests

Instead of selling the token, the investor pledges it as collateral and receives a loan. The vault or LP capital pool becomes a secured lender rather than a buyer, earning interest and fees from the loan.

The basic structure is:

1. The investor owns a compliant tokenized interest.
2. The investor pledges the token into a collateral contract or controlled account.
3. The vault lends a percentage of NAV, subject to a haircut.
4. The investor continues to own the economic exposure unless default occurs.
5. Interest accrues to the LP lenders.
6. If collateral coverage deteriorates, the protocol triggers margin calls, partial liquidation, forced sale to the vault, or wind-down enforcement.

This structure is closer to margin lending, repo, private credit facilities, and NAV lending than to a redemption program.

6.3 Pathway 3: Hybrid Liquidity

Hybrid structures combine purchase and credit. The investor may receive a loan with an embedded right for the vault to buy collateral if coverage falls. The vault may advance liquidity as debt but convert to ownership at a predefined discount on default. LPs can allocate capital between buyer tranches and lender tranches. The protocol can offer partial sale plus partial loan, reducing forced selling pressure.

Hybrid liquidity is especially useful where outright exit would impose an excessive discount but the investor only needs temporary liquidity.

6.4 Pathway 4: Bridge and Asset-Level Finance

The same vault can provide asset-level credit, subject to legal and governance constraints. For real estate, this could include renovation finance, tenant-improvement finance, construction bridge capital, or temporary working-capital support. For infrastructure, it could include milestone bridge financing or capex support. For private credit, it could include warehouse-style liquidity against expected collections. For royalty or lease assets, it could include advances against contracted cash flows.

This function should be treated carefully because asset-level finance can change the risk profile for all token holders. It requires disclosed borrowing authority, use-of-proceeds restrictions, collateral and seniority rules, draw controls, governance approval thresholds, covenant reporting, and default procedures.

6.5 Collateral Haircuts

Borrowing capacity should be more conservative than exit pricing. If a vault might buy an asset at NAV less a 7 percent liquidity discount, it should lend against that asset at a deeper haircut to account for enforcement delay and price uncertainty.

Illustrative loan-to-value ranges are shown in Table 2. These are not proposed offering terms. They illustrate that ownership-backed credit must price asset class, collateral enforceability, NAV freshness, and recovery horizon.

Table 2: Illustrative Ownership-Backed Loan-to-Value Ranges

Asset Type	Illustrative LTV Range
Stabilized real estate token	40–60%
Diversified private-credit pool	50–70%
Infrastructure project interest	35–60%
Royalties or IP cash flows	25–50%
Litigation finance exposure	10–35%
Natural-resource concession	20–50%

7 Real Estate as Phase 1 Implementation

Real estate is not the thesis, but it is the most practical first implementation. It is familiar to investors, legally legible, cash-flowing, collateralized, and structurally illiquid. A real-estate implementation can test all six universal primitives: tokenized ownership, asset-specific vaults, NAV oracles, dynamic discounts and haircuts, compliance controls, and wind-down.

7.1 Why Real Estate First

Real estate has several properties that make it an appropriate Phase 1 market:

- the asset is tangible and commonly understood;
- valuation through appraisal is imperfect but familiar;
- rental income creates observable cash-flow data;
- title and transfer systems create enforceable legal anchors;
- investors already understand debt, equity, refinancing, and liquidation;
- the market is visibly illiquid, making the problem easy to explain.

7.2 Kenya Diaspora Proof of Concept

Kenya remains useful as a constrained proof of concept. Formal diaspora remittances to Kenya reached USD 4.945 billion in 2024, up from USD 4.190 billion in 2023 according to Commonwealth Secretariat reporting citing Central Bank of Kenya data [26, 27]. This figure is not itself addressable investment volume; it is a corridor indicator. The relevant hypothesis is that a small share of discretionary remittance capital could be redirected into regulated, asset-backed investment if the structure offers credible ownership, compliance, liquidity, credit, and recovery.

The Kenya PoC should be framed as a narrow validation environment:

- tokenize 1–2 real-estate assets or SPV interests;
- attach an asset-specific liquidity and credit vault;
- test availability fees, secured borrowing, and discounted exits;
- validate NAV update cadence and staleness penalties;
- test compliance credentials across investor-origin jurisdictions;
- disclose that liquidity is conditional, asset-specific, and not guaranteed.

7.3 Gross-to-Net Cash Flow

Investor distributions would be based on net operating income, not headline rent. For a representative Nairobi satellite-town asset, the following baseline is illustrative and must be replaced with actual property data before any offering:

Table 3: Illustrative Real-Estate Gross-to-Net Yield Waterfall

Component	Assumption	Note
Gross Rental Yield	9.0%	Market baseline to verify [28]
Property Management Fee	-1.0%	Paid to local operator
Vacancy & Arrears Reserve	-1.5%	Buffered quarterly
Maintenance & Insurance	-1.0%	Escrowed monthly
Platform & Protocol Fees	-1.0%	Compliance, oracle, operations
FX Conversion	-0.5%	KES to stablecoin conversion friction
Net Distributable Yield	~4.0%	Paid pro-rata to holders

The vault economics are separate from the asset distribution waterfall. LPs may earn availability fees, loan interest, and discounted acquisition upside; token holders may earn asset-level distributions. Where the vault acquires ownership positions, LPs may also receive the asset-level yield attached to those positions.

8 Comparison Against Existing Models

Table 4 compares SETTLEY against existing structures, focusing on liquidity, redemption, collateralization, and recovery rather than tokenization alone.

Table 4: Comparison Against Existing Market Structures

Model	Liquidity	Redemption	Collateralization	Recovery	Settley Difference
Public REITs	High exchange liquidity	Investor sells shares; no asset-level redemption	Shares may be marginable	Corporate and securities-law process	Asset-level conditional liquidity and credit, not listed portfolio exposure
Non-traded REITs	Limited, programmatic, often capped	Repurchase programs may be limited or suspended	Usually no direct investor borrowing against asset-level ownership	Manager-led asset sale or fund process	Facility size, utilization, discounts, haircuts, and availability explicit at asset level
RealT	Tokenized fractional real estate; liquidity depends on market depth	No guaranteed redemption	Limited direct protocol-native lending	Property and legal-wrapper recovery	Vault as liquidity and credit balance sheet, not only a transfer venue [4]
Lofty	Fractional property marketplace with platform-enabled sale mechanics	Liquidity depends on buyers and platform rules	Limited standardized collateral use	Property sale or platform process	Issuance separated from explicit LP-funded liquidity, lending, and discounted acquisition [5]
Private equity funds	Fund interests are illiquid; secondaries exist	Redemptions usually unavailable or restricted	NAV lending exists mainly at fund or sponsor level	Fund documents and portfolio sale	NAV-based liquidity and credit pushed to tokenized ownership positions
Tokenized securities	Transferability improves settlement	Redemption depends on issuer or product terms	Collateral use varies	Legal wrapper and securities process	Tokenization treated as representation, not liquidity
Private credit funds	Periodic liquidity, gates, or closed-end structures	Manager-controlled subscriptions and redemptions	Facilities secured by loans or NAV	Loan collections and collateral enforcement	Loan-pool-specific liquidity, credit, and collateral rules
Repo markets	Deep for high-quality securities	Repurchase, not redemption	Core design is collateralized lending	Collateral sale on default	Repo logic adapted to illiquid RWA tokens with larger haircuts and slower recovery [11]
NAV lending	Liquidity against portfolio NAV	Borrower repays facility; not investor redemption	Secured by fund or portfolio interests	Lender rights over fund assets or cash flows	NAV-backed lending generalized to asset-specific tokenized ownership

9 Compliance and Legal Architecture

SETTLEY is proposed infrastructure, not a claim that code alone can create legal ownership or regulatory compliance. Tokenized interests conferring income, profit participation, or governance rights may be securities, fund interests, lending products, commodities-linked instruments, or other regulated instruments depending on jurisdiction and structure. In the United States, the tokenized form of a security does not remove the application of securities laws [16, 17]. In Europe, the MiCA regime and existing securities frameworks must be evaluated by product type [18]. UK financial promotions and cryptoasset rules may constrain marketing and distribution [19].

The legal architecture should separate:

- the legal person or governing entity;
- the asset-holding vehicle or asset pool;
- the token registry;
- the liquidity and credit vault;
- the compliance credential layer;
- the off-chain valuation, servicing, and recovery process.

For the Phase 1 real-estate PoC, a Wyoming DAO LLC or similar recognized legal entity may own or control jurisdiction-specific SPV interests, while smart contracts record balances, enforce transfer restrictions, manage vault operations, and automate disclosed thresholds [20, 21]. In Kenya, property-holding SPVs would need to be assessed under the Companies Act, Capital Markets Act, land-registry rules, data-protection law, and CMA sandbox requirements [22, 23, 24, 25].

10 Risk and Residual Failure Modes

SETTLEY reduces some forms of opacity, but it does not eliminate risk. The protocol must disclose that liquidity is conditional, credit is collateral-dependent, and recovery can be slow or impaired.

Table 5: Risk Matrix

Risk Vector	Mitigation	Residual Risk
Vault depletion	Asset-specific facility limits, utilization caps, availability disclosures, circuit breakers	Facility capacity can be exhausted; exits and loans may pause
Collateral impairment	Conservative LTVs, NAV haircuts, monitoring, margin calls	Collateral may fall faster than enforcement can occur
NAV staleness	Scheduled valuation, staleness penalties, emergency impairment workflow	NAV may still be wrong between updates
LP concentration	Asset-specific disclosure, tranching, facility caps	LPs may over-concentrate in weak assets or correlated exposures
Borrower default	Collateral pledge, margin call, liquidation rights	Enforcement may be delayed by law, courts, or operational failure
Title or ownership defect	Legal diligence, registry checks, legal opinions, insurance where available	Fraud, historical claims, or senior claims may impair recovery
Regulatory reclassification	Jurisdiction screening, credential gating, legal opinions, sandbox engagement	Regulators may restrict issuance, lending, transfer, or marketing
Operator or servicer failure	Replacement rights, audits, segregated accounts, covenants	Cash-flow disruption may impair distributions and loan repayment
Smart-contract exploit	Audits, formal verification for critical functions, pause controls	Bugs, key compromise, and integration failures remain possible

11 Implementation Roadmap

11.1 Phase 1: Conditional Liquidity and Credit for Real Estate

The first phase validates the architecture in a narrow real-estate setting. The goal is not broad retail distribution. The goal is to test whether an asset-specific vault can support availability fees, ownership-backed loans, discounted exits, compliance-gated transfers, NAV updates, and wind-down logic under real legal constraints.

11.2 Phase 2: Conditional Liquidity and Credit for Illiquid RWAs

After a real-estate proof point, the same architecture can expand to infrastructure assets, private credit pools, natural-resource interests, and alternative cash-flowing assets. The universal components remain constant; valuation, legal wrapper, cash-flow servicing, and recovery modules change.

11.3 Phase 3: Ownership-Backed Credit Markets

SETTLEY evolves from exit liquidity to ownership-backed credit. Token holders can pledge ownership interests as collateral. LPs can become facility providers, lenders, buyers, or hybrid capital providers. This creates a programmable credit market around illiquid ownership.

11.4 Phase 4: Global Infrastructure Ownership and Financing Layer

The long-term vision is a global market structure where illiquid ownership interests can be issued, financed, transferred, lent against, improved, refinanced, and recovered through standardized but asset-specific liquidity and credit infrastructure.

12 Limitations

This paper is a protocol and legal-architecture proposal. It does not establish product-market fit, regulatory approval, enforceability in any specific jurisdiction, realized investor demand, LP appetite, borrower demand, or the performance of any asset. The liquidity model, availability fees, reserve or facility sizing, loan-to-value ratios, acquisition discounts, utilization thresholds, tranche economics, and stress assumptions are illustrative. They are not forecasts, promised returns, or offering terms.

Before any live offering, the structure requires legal opinions, regulatory engagement, asset-specific underwriting, audited smart contracts, regulated onboarding partners, custody arrangements, tax analysis, and pilot data validating investor demand, LP appetite, loan behavior, exit behavior, valuation accuracy, and operating costs.

13 Conclusion

SETTLEY is not a real-estate protocol. It is conditional liquidity infrastructure for illiquid assets.

The distinction matters. Tokenization can make ownership programmable, but programmable ownership is not the same as liquidity. Liquidity requires funded balance sheets, collateral rules, valuation governance, compliance constraints, and recovery mechanisms. SETTLEY proposes to supply those missing layers through asset-specific vaults that function as programmable credit facilities.

Real estate is the first implementation because it is tangible, cash-flowing, and visibly illiquid. But the architecture is broader. Infrastructure projects, private credit pools, natural-resource rights, royalty streams, equipment leases, and other alternative assets face the same structural problem: ownership may be valuable, but exit and credit are conditional.

SETTLEY's design does not pretend to eliminate illiquidity. It makes illiquidity explicit. It prices immediacy through discounts. It prices borrowing through haircuts. It compensates LPs through availability fees, loan interest, discounted acquisition upside, and asset-level yield where applicable. It defines what happens when liquidity fails.

That is the paper's core claim: the next frontier of RWA tokenization is not issuance. It is the creation of conditional liquidity and ownership-backed credit markets for assets that were never naturally liquid in the first place.

References

- [1] Savills World Research. *Total Value of Global Real Estate*. Estimate of global real-estate value at USD 300–380 trillion.
- [2] U.S. Securities and Exchange Commission. *Real Estate Investment Trusts (REITs): Investor Bulletin*. <https://www.sec.gov/resources-for-investors/investor-alerts-bulletins/reits>
- [3] Financial Industry Regulatory Authority. *Non-Traded REITs: Real Estate Investment Trusts*. <https://www.finra.org/investors/insights/non-traded-reits-real-estate-investment-trusts>
- [4] RealT. *Fractional U.S. rental-property tokens*. <https://realt.co/>
- [5] Lofty. *Fractional real estate marketplace*. <https://www.lofty.ai/>
- [6] OECD. *Infrastructure Financing Instruments and Incentives*. OECD, 2015.
- [7] World Bank. *Private Participation in Infrastructure and Infrastructure Finance Materials*. <https://ppi.worldbank.org/>
- [8] OECD. *The Tokenisation of Assets and Potential Implications for Financial Markets*. OECD Blockchain Policy Series, 2020.
- [9] Bank for International Settlements. *Annual Economic Report 2023*, Chapter III: Blueprint for the Future Monetary System. <https://www.bis.org/publ/arpdf/ar2023e3.htm>

- [10] ERC-3643 Association. *ERC-3643: Permissioned Token Standard for RWA Tokenization*. <https://www.erc3643.org/>
- [11] Federal Reserve Bank of New York. *Repo and Reverse Repo Agreements*. https://www.newyorkfed.org/markets/rrp_faq
- [12] Financial Industry Regulatory Authority. *Margin Accounts*. <https://www.finra.org/investors/investing/investment-products/stocks/margin-accounts>
- [13] Weyl, E. G., Ohlhaver, P., and Buterin, V. *Decentralized Society: Finding Web3's Soul*. SSRN Working Paper, 2022. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4105763
- [14] ERC-5192: Minimal Soulbound NFTs. Ethereum Improvement Proposals. <https://eips.ethereum.org/EIPS/eip-5192>
- [15] World Wide Web Consortium. *Verifiable Credentials Data Model*. W3C Recommendation. <https://www.w3.org/TR/vc-data-model/>
- [16] U.S. Securities and Exchange Commission. Statement on tokenized securities: the tokenized format of a security does not change the application of the federal securities laws.
- [17] *SEC v. W. J. Howey Co.*, 328 U.S. 293 (1946).
- [18] European Union. *Regulation (EU) 2023/1114 on Markets in Crypto-Assets (MiCA)*, OJ L 150, 9 June 2023.
- [19] Financial Conduct Authority. *Guidance on Cryptoassets* (PS19/22), 2019; and *Cryptoasset Financial Promotions Regime* (PS23/6), 2023.
- [20] State of Wyoming. *Wyoming Decentralized Autonomous Organization Supplement*, W.S. 17-31-101 et seq.
- [21] Wyoming Secretary of State. *Decentralized Autonomous Organization (DAO): Frequently Asked Questions*. https://sos.wyo.gov/Business/Docs/DAOs_FAQs.pdf
- [22] Republic of Kenya. *Companies Act, No. 17 of 2015*. <http://kenyalaw.org/>
- [23] Republic of Kenya. *Capital Markets Act*, Cap. 485A, Laws of Kenya. <http://kenyalaw.org/>
- [24] Capital Markets Authority (Kenya). *Regulatory Sandbox Policy Guidance Note*, 2019. <https://www.cma.or.ke/>
- [25] Republic of Kenya. *Data Protection Act, No. 24 of 2019*. <http://kenyalaw.org/>
- [26] Commonwealth Secretariat. *Kenya's Diaspora Investment Strategy*, citing Central Bank of Kenya, 2025.
- [27] Central Bank of Kenya. *Diaspora Remittances*. <https://www.centralbank.go.ke/diaspora-remittances/>
- [28] Vasilisa Africa. *Nairobi Residential Yields Report*, 2025. Founder-supplied source; to be independently verified.